Fisheries Resource Center of Indonesia (FRCI)

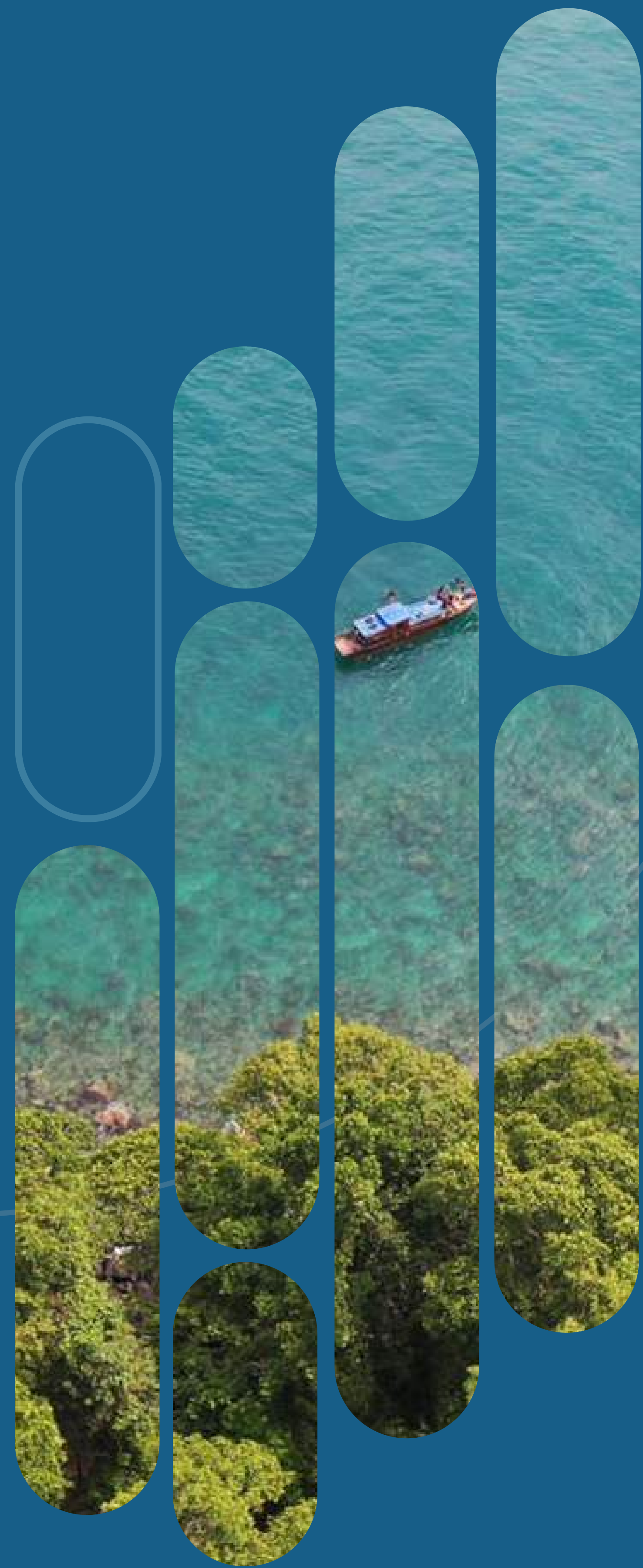
Jl. Sempur No.35, RT.03/RW.01, Sempur, Kecamatan Bogor Tengah, Kota Bogor, Jawa Barat 16129

perikanan.org

FRCI
FISHERIES RESOURCE CENTER OF INDONESIA

IMPACT REPORT 2024

FISHERIES RESOURCE CENTER OF INDONESIA



REKAM NUSANTARA FOUNDATION

FRCI
FISHERIES RESOURCE CENTER OF INDONESIA

FISHERIES RESOURCE CENTER OF INDONESIA

IMPACT REPORT 2024

Who we are

The Fisheries Resource Center of Indonesia (FRCI) was initiated after many discussions with fishery experts from universities and organizations who recognize the importance of data management as the basis for formulating public policies.

As one of the working units under the Rekam Nusantara Foundation, FRCI was established to provide an alternative to fisheries analysis and management of sustainable marine resources based on scientific data with the vision of

"Sustainability and Justice for Indonesian Fisheries".

TABLE OF CONTENTS

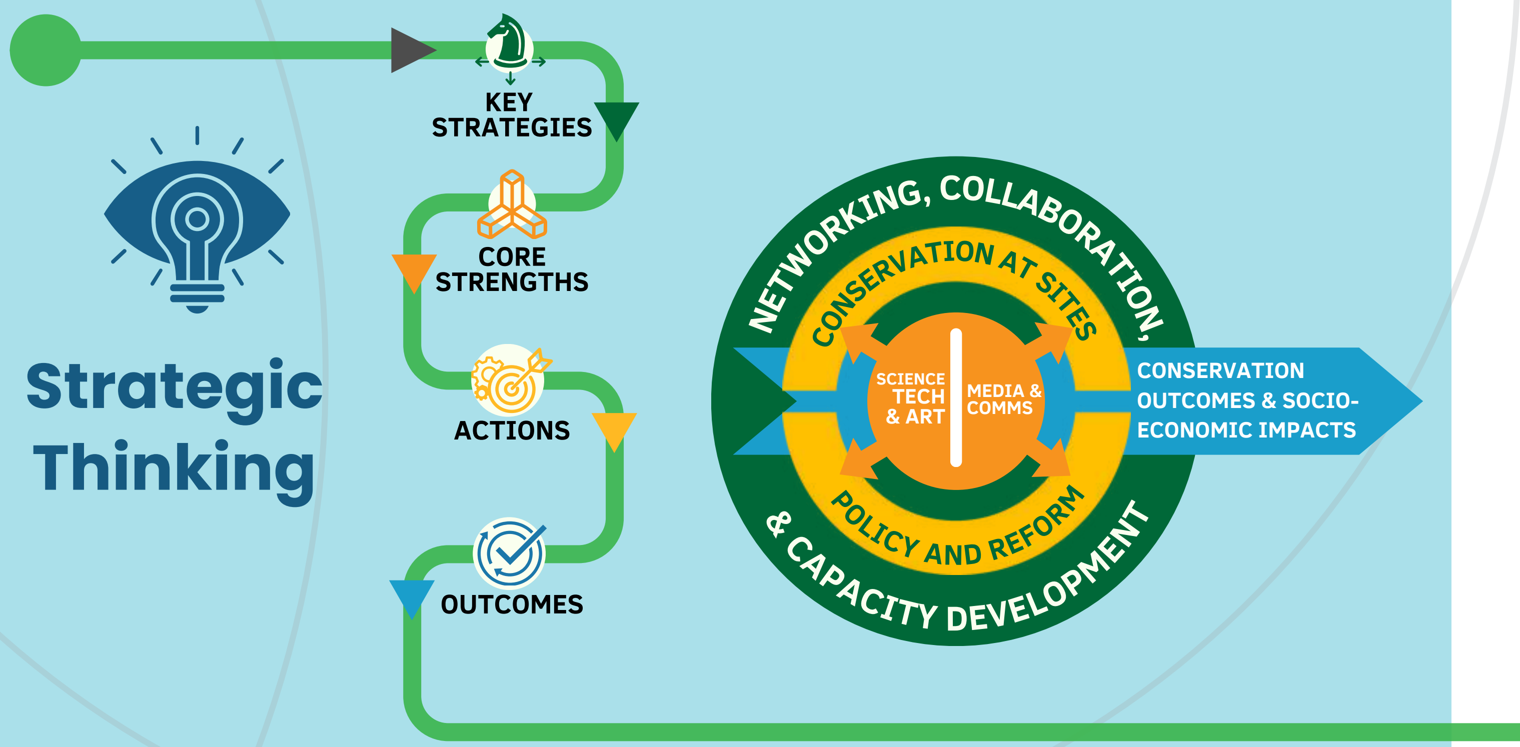
FOREWORD FROM THE CHAIRPERSON

When we founded the Rekam Nusantara Foundation, our mission was clear: to share knowledge about Indonesia's incredible natural richness. This includes not just its biodiversity but also the deep connection between people, culture, and the environment. Through knowledge and art, we tell stories about Indonesia's landscapes, its people, and the way they interact with nature. We've journeyed across the country—trekking through jungles, scaling mountains, and diving into the ocean—to capture these stories. We call this ***#merekamnusantara***. By sharing what we discover, we hope to inspire more people to appreciate and protect Indonesia's natural treasures.

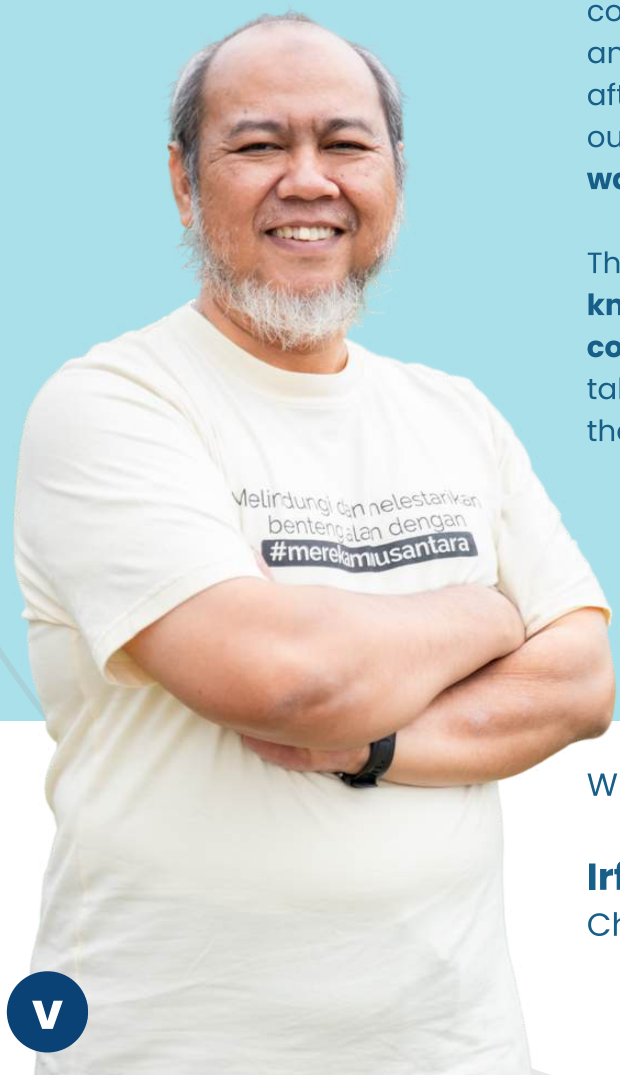
But along the way, we realized that raising awareness alone wasn't enough. Real conservation requires collective action. That's why we expanded our efforts—working alongside communities, government agencies, universities, businesses, and other partners to find sustainable solutions. Five years after our establishment in 2013, we took a step back to refine our approach, asking ourselves: **What is the most effective way to protect Indonesia's natural resources?**

The answer became clear. We needed to **bridge scientific knowledge with technology, local wisdom, and creative communication to drive real impact.** This means not only taking direct conservation action but also shaping policies that support long-term sustainability.

With this in mind, I'm proud to introduce Rekam Nusantara Foundation's new approach—built on our core strengths, key strategies, and a clear vision for the future. By weaving these elements together, we aim to create lasting conservation outcomes that benefit both nature and people. Our goal is simple: **a future where humans and nature thrive together, just as the founders of our archipelago envisioned**.



"We recognized the need to integrate scientific knowledge, technology, local wisdom, and creative communication to drive conservation efforts that create meaningful, lasting benefits for both nature and society."



With gratitude,

Irfan Yulianto
Chairperson, Rekam Nusantara Foundation

FOREWORD FROM THE DIRECTOR

With great pride and a deep sense of responsibility, we present the annual Impact Report of the Fisheries Resource Center of Indonesia (FRCI). This report summarized the significant strides we have made in advancing marine conservation and sustainable fisheries management, as well as the invaluable contributions of our dedicated team, partners, and communities.

Over the past year, FRCI has continued to push the boundaries of what is possible in the realm of marine conservation. Our flagship programs, i.e., sustainable fisheries, ocean accounts, species conservation, marine conservation, and blue carbon, focused on **empowering local coastal communities, advancing science-based management practices, and ensuring that Indonesia's rich marine biodiversity is preserved for future generations.** Through partnerships with local governments, academics, civil society organizations, and community groups, we have implemented effective conservation strategies and enhanced livelihoods for coastal communities.

One of our key achievements was co-organized The Global Dialogue on Sustainable Ocean Development with the Ministry of Marine Affairs and Fisheries and the Global Ocean Accounts Partnership (GOAP) in Bali. This important event brought community representatives, technical experts, senior decision-makers, and researchers to strengthen the global ocean accounting community of practice. In addition, our work at the site level has successfully expanded the collaborative initiatives with 11 coastal communities across Indonesia. These partnerships have resulted in the development of sustainable fisheries management programs, the enhancement of marine protected areas (MPAs), and the establishment of documents used for marine protected areas

This report also highlights the ongoing support we receive from donors and stakeholders, whose contributions are critical to the success of our work. Without their commitment, none of our achievements would have been realized. Looking ahead, we are keenly aware that the challenges we face are immense, but we are also optimistic about the opportunities for progress. Through collaboration, innovation, and perseverance, we are confident we can continue to make a meaningful impact. Together, we can secure a future where people and nature can thrive in harmony.

With sincere appreciation

Heidi Retnoningtyas
Director, Fisheries Resource Center of Indonesia
Rekam Nusantara Foundation

FRCI by the numbers

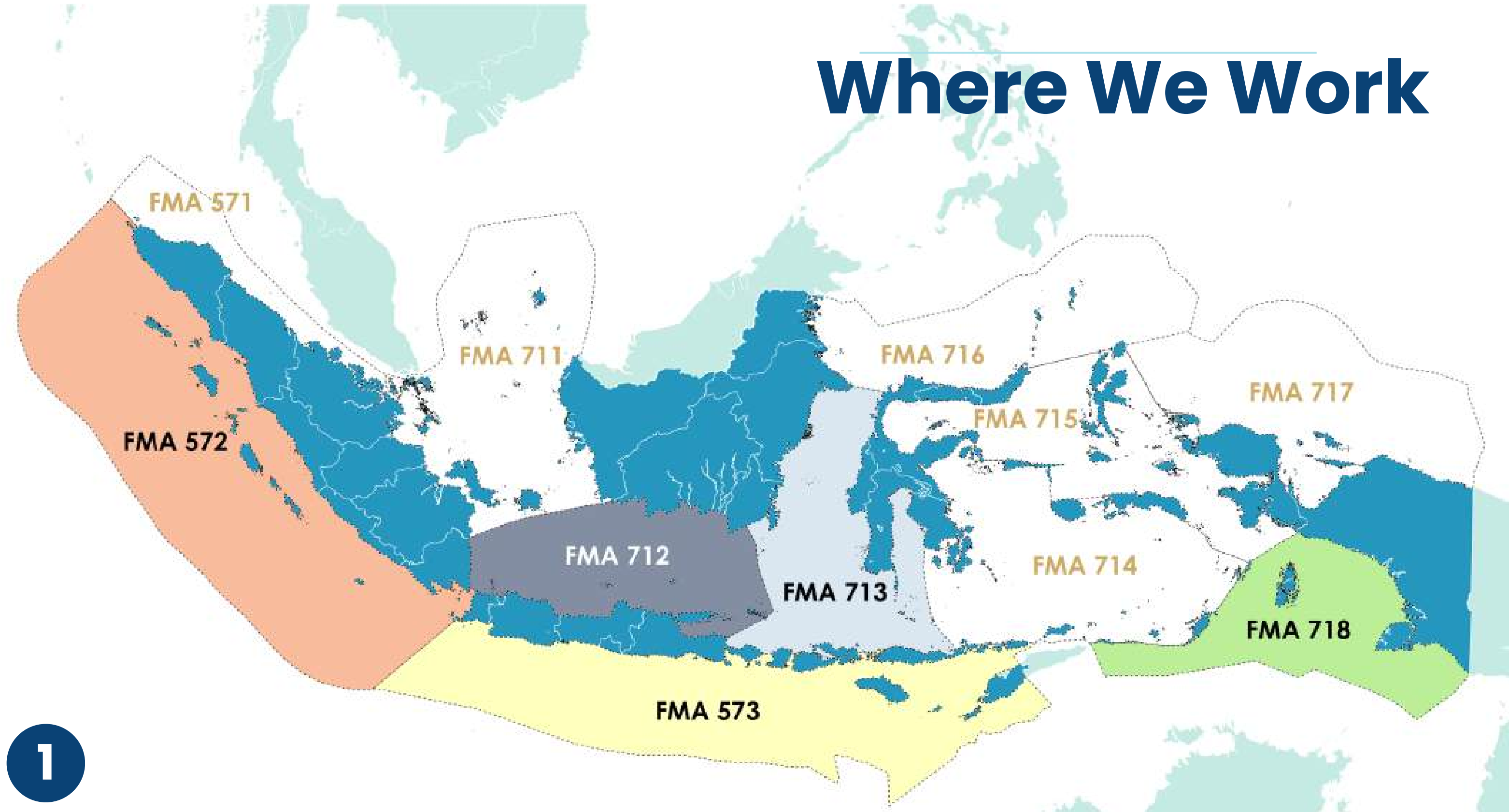
2022 - 2024

- **16** Marine Protected Areas (MPAs)
- **5** Area-Based Management
- **5** Fisheries Management Areas (FMAs)

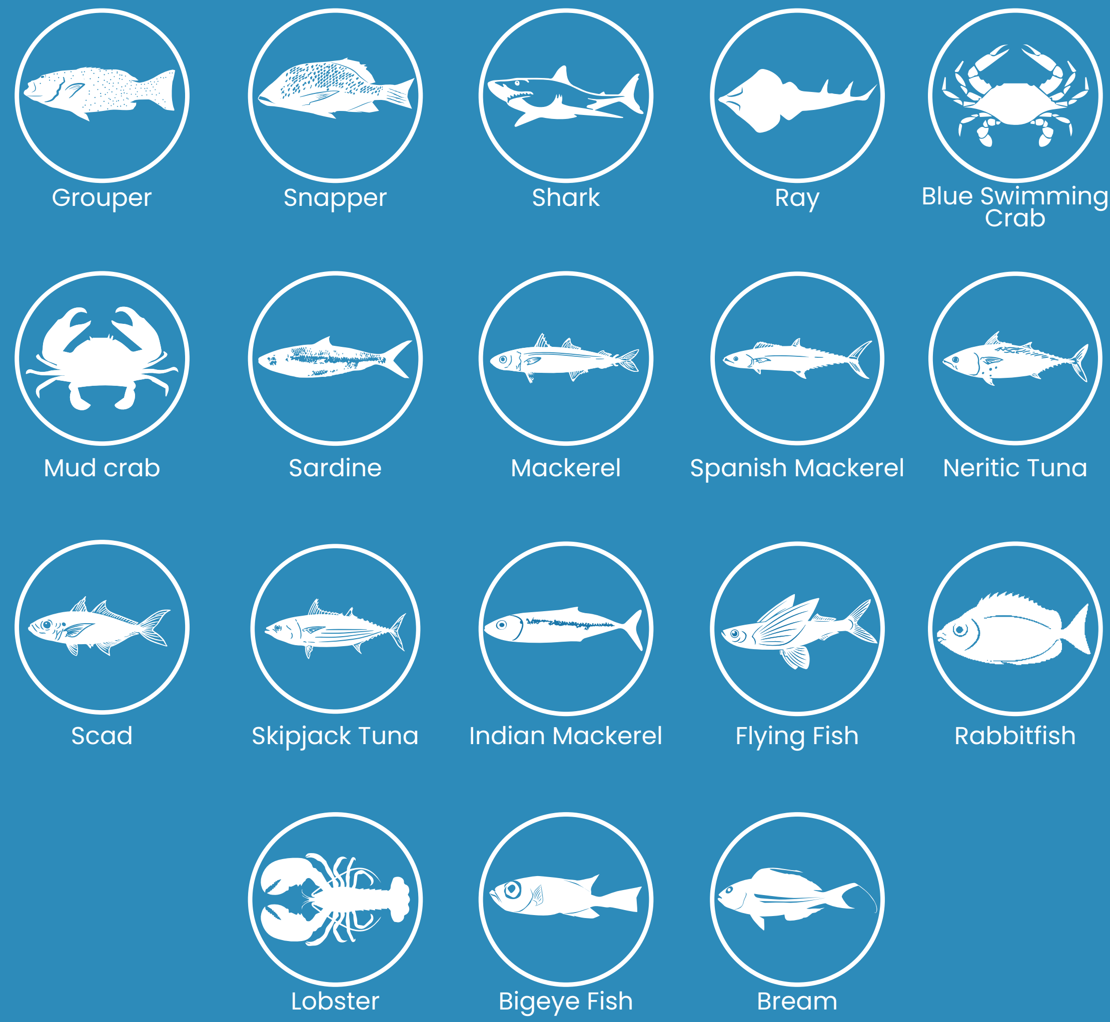
2024

- **11** Community Groups
- **30** Field Enumerators
- **7** Student Internships
- **7** Volunteers
- **9** Journal Publications

Where We Work



Fisheries Commodities



23,090

fish length or carapace width data collected in 2024

Ecosystem Based Fisheries Management (EBFM)

To improve the **management of small-scale fisheries** in Saleh Bay, we are conducting scientific research integrated with policy through an Ecosystem-Based Fisheries Management (EBFM) study.

EBFM in Saleh Bay is an approach that **integrates ecosystem indicators into fisheries management** to ensure sustainable resource utilization.

TOOLS: The **Ecopath with Ecosim (EwE)** model, which can be used to understand key species and ecosystems and their interactions.

What we do

1. Held training workshop to improve the capacity related to EwE and EBFM
2. Develop the structure and collate data sets for an (EwE) model
3. Held stakeholder meetings at the provincial and national level.

Data Collection

42 Desktop Study (Functional Groups Identified)

90 Diet Compositio Study (Samples from 16 Functional Groups)

210 Socio Economic Survey (Respondents)

72 Socio Economic Survey (Key Informations)

15 Socio Economic Survey (Vilages)

Socio Economic Survey

210 respondents

72 key informations

15 vilages

SNAPPER & GROUPER

National Snapper-Grouper Initiative

We worked with YKAN to conduct stock assessments for Snapper and Grouper, initiated as a scientific service provider (SSP) for the National Commission on Fish Stock Assessment (KOMNAS KAJISKAN)

The Fishery Improvement Project (FIP) in Saleh Bay, West Nusa Tenggara

is focused on the sustainability of snapper and grouper fisheries

The goal is to develop and implement strategies for:

- Stock rebuilding
- Harvest control rules
- Manage habitat and ecosystem

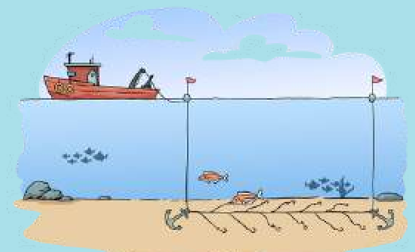
UNIT OF ASSESSMENT (UOA)

Species

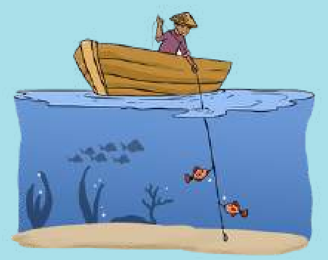
Malabar blood snapper
(*Lutjanus malabaricus*)

Leopard grouper
(*Plectropomus leopardus*)

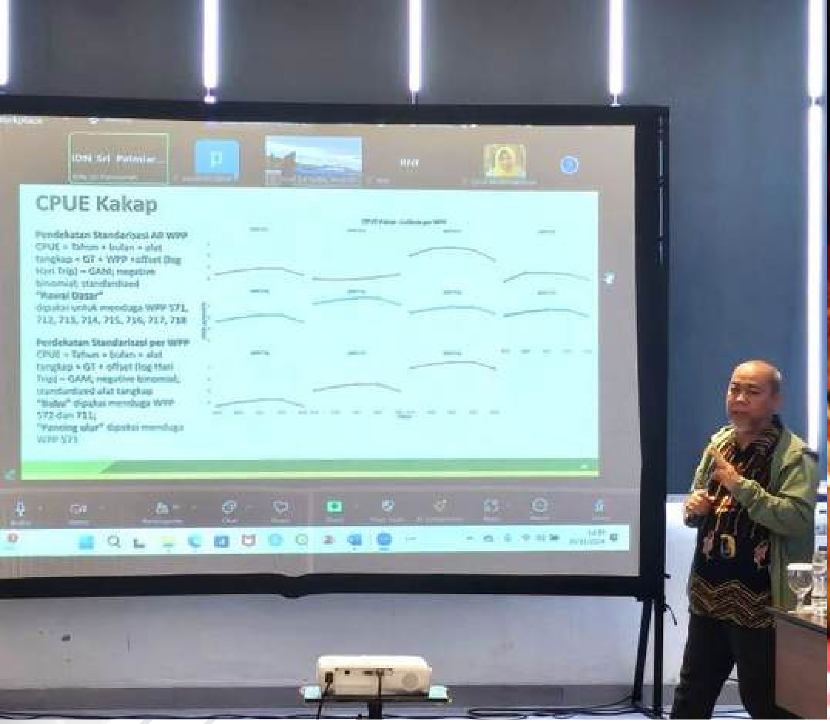
Fishing gear



Bottom longline



Handline



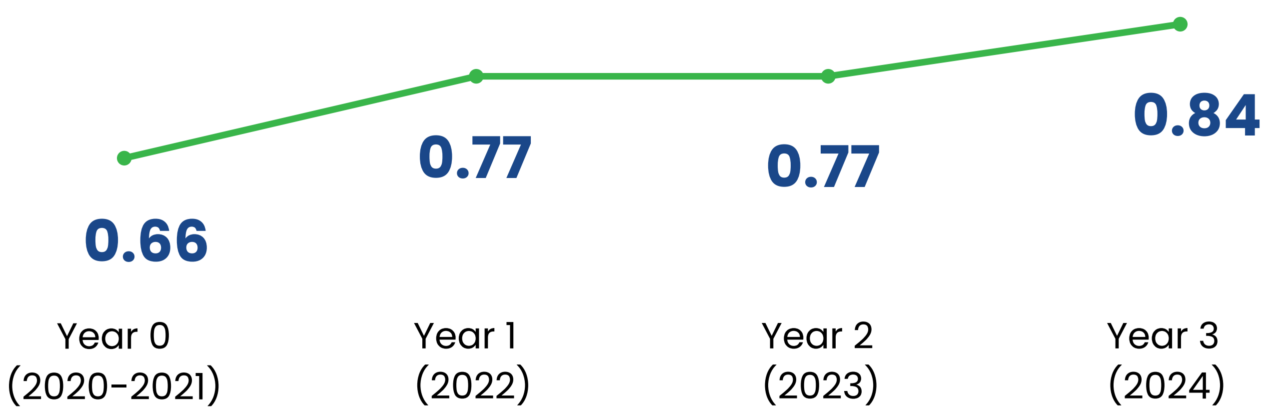
FIP at Stage 4

Our improvements in Fishing Practices or Fishery Management and Progress received an A (Advanced Progress) rating from FisheryProgress.org

The current score of the BMT (Benchmarking and Tracking Tool) index is 0.84

- 75% for Green indicator
- 18% for Yellow indicator
- 7% for Red indicator

BMT Progress Tracker



PELAGIC FISH of FMA 572

We conducted a framework survey in collaboration with local partners to understand the condition of each fish landing site and the distribution of the fishing fleet in fisheries management area (FMA) 572.

COLLECTED INFORMATION:

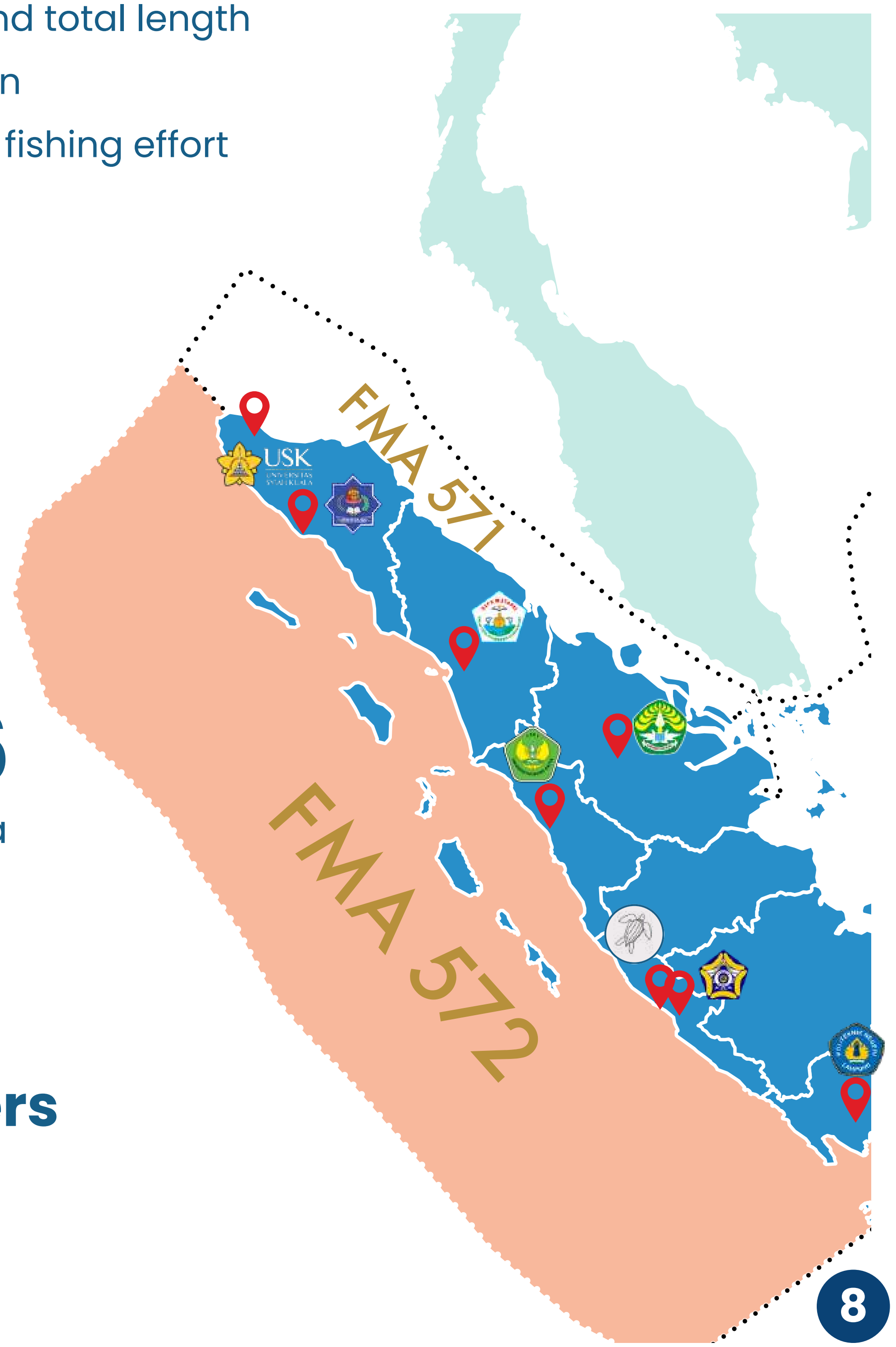
1. Fish production and total length
2. Catch composition
3. Characteristics of fishing effort

12 sites

1,839 trips

8,816 length data

8 Local partners



CRAB INITIATIVES

We conducted catch monitoring of Blue swimming crab and Mangrove mud crabs to understand the intricate relationship between these species and the mangrove ecosystem in 5 locations.

What we do

Data collection using IKAN

Fishing trip activities

Catch data

Biological data

Photo samples

7 Volunteers

7 Enumerators

participated in data collection

which collected

11,265

biological data of blue swimming crab and mud crab

Commodities

Blue swimming crab (*Portunus pelagicus*)

Giant mud crab (*Scylla serrata*)

Orange mud crab (*Scylla olivacea*)

Outputs

- Stock status of Blue swimming crab in Demak, Central Java
- Stock status of Blue swimming crab in Jepara, Central Java
- Stock status of Blue swimming crab in Saleh Bay, West Nusa Tenggara
- Stock status of Giant mud crab in Demak, Central Java

Marine Protected Area (MPA) Management

Our work:
Support the Indonesian national and local governments in strengthening effective, adaptive and sustainable management of conservation areas to benefit ecosystems and communities.

Approach:

- Strengthening government policies in conservation area management
- Encouraging active community participation in conservation area management
- Improving the institutional capacity and human resources of conservation area managers

1. National: MPA Vision 2045

By 2045, Indonesia aims to expand the coverage of MPAs to **30% of the total marine area**, or at least **97.5 million hectares.**

What we do:

- Supported in spatial analysis to identify potential areas for extending existing MPAs and proposing new MPAs
- Supported in developing a strategy document for MPA Vision 2045

Output

What's Next :

- Finalising strategy document of MPA Vision 2045
- Launching of MPA Vision 2045

2. National: Offshore MPA

Pilot Project: Offshore MPA in Sulawesi Sea

Aims:
Develop MPA for offshore fisheries as one of the strategies for expanding the coverage of MPAs to 97.5 million hectares and to promote sustainable fisheries.

What we do:

- Supported an analysis for zoning plan on the dispersal of pelagic fish larvae
- Supported in developing a Management Plan Document of the first Indonesia's MP
- Supported in developing the first guidelines for the establishment and management of MPA

Outputs:

- Information on issues and challenges in MPA for offshore fisheries management
- Draft of MPA for offshore fisheries guidelines

3. Liukang Tangaya MPA

Our main programs:
- Support effective and adaptive governance of Liukang Tangaya MPA
- Reduce the threat of destructive fishing practices
- Increase community participation and support in conservation area management

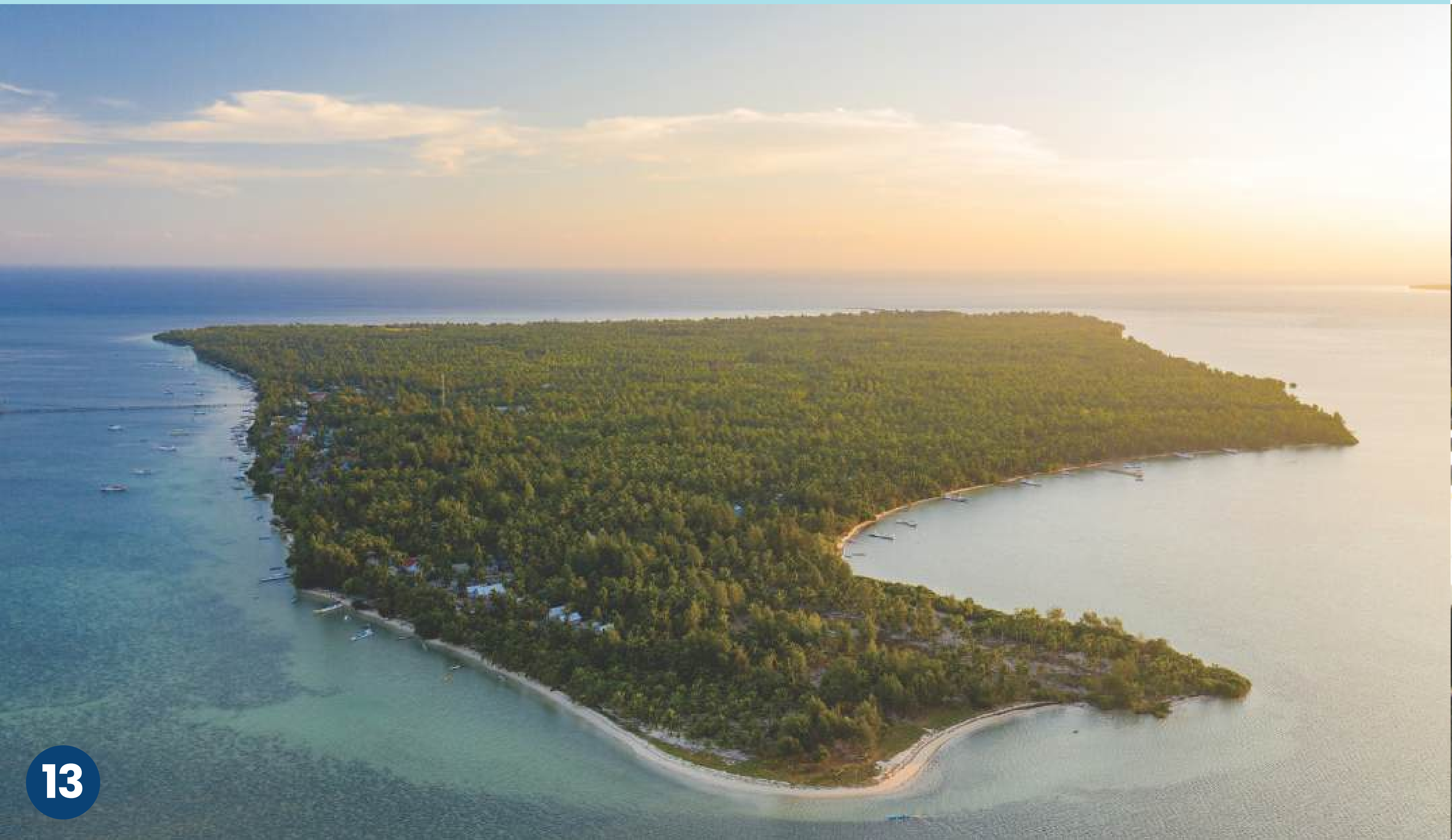
Achievement:
- Supporting the establishment of the Liukang Tangaya MPA Management Unit
- Socialization of MPA management to village governments and communities to more **300 people**
- Final draft of Management Plan of Liukang Tangaya MPA
- **MPA 101 training and certification** for 3 Liukang Tangaya MPA managers.

EVIKA (Evaluasi Efektivitas Pengelolaan Kawasan Konservasi) is Indonesia's national evaluation tool designed to assess the management effectiveness of Marine Protected Areas (MPAs). This tool aims to provide a comprehensive assessment of MPA management, facilitating targeted improvements and ensuring the sustainability of marine conservation efforts.

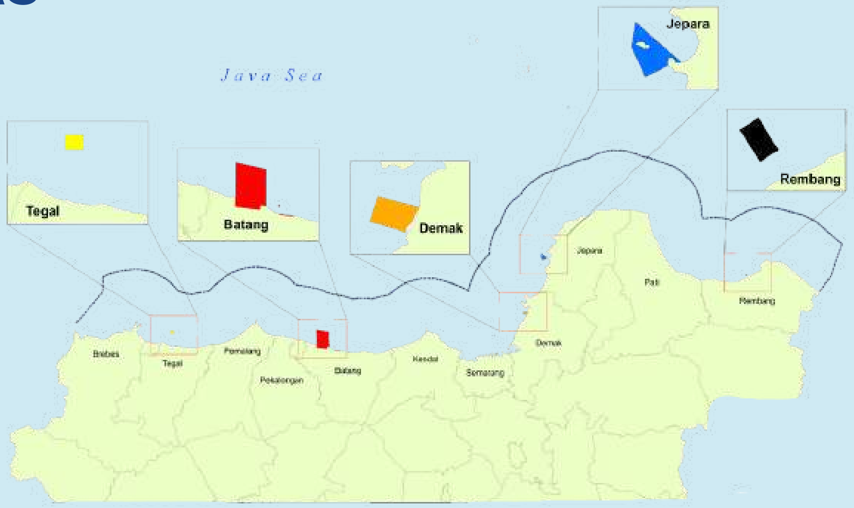
Liukang Tangaya MPA EVIKA Score

2022	2023	2024
Assigned	21.67%	41%

Liukang Tangaya MPA scores in the minimum category (<50%). However, there is a good improvement each year.



4. Central Java MPAs



What we do

- Regulatory and policy support
- Empowerment of local communities
- Ecosystem conservation and rehabilitation
- Monitoring of marine protected areas
- Awareness, research, and education

Outputs

- Socio-economic Report
- Ecological Data on Mangrove Seagrass Ecosystem
- Data and Report on Ocean Accounts
- Campaign and awareness of MPA to more than 300 communities
- MPA 101 training and certification for 20 Central Java MPA managers

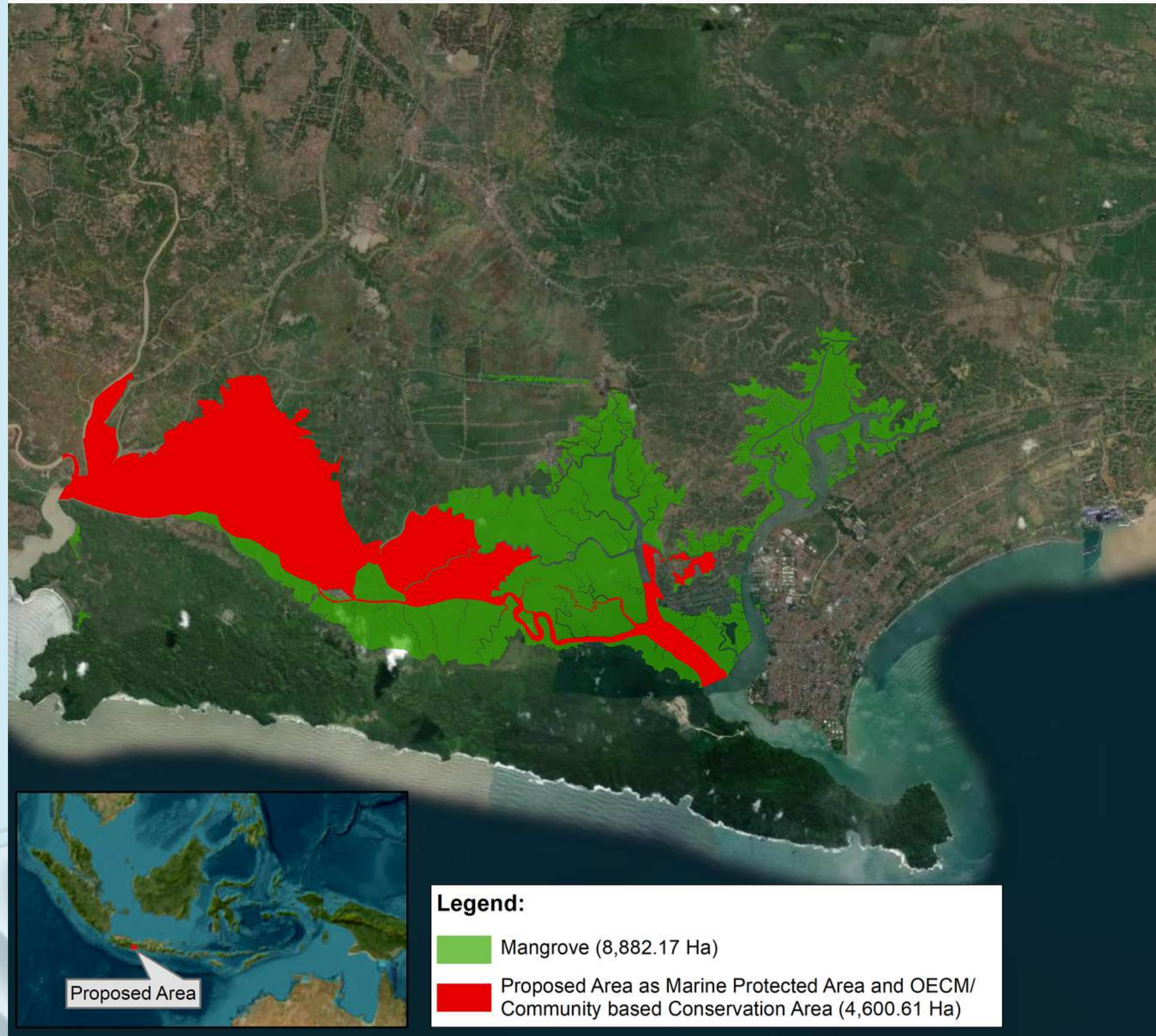
EVIKA scores

	2022	2023	2024
Ujungnegoro Roban	40.54 %	43.06 %	49.28 %
Pulau Panjang	Assigned	13.95 %	41.51 %
Karang Jeruk	Assigned	14.53 %	46.98 %
Karang Jahe	Assigned	13.94 %	45.13 %
Betahwalang	-	-	Assigned

Central Java MPAs scores in the minimum category (<50%). But there is an increase every year.

5. Segara Anakan Lagoon

We facilitate dialogues between national and provincial stakeholders about the initiative of new conservation areas in Central Java. This process demonstrated open communication among stakeholders about the plan to designate and manage **Segara Anakan as a national specific strategic area.**



BLUE CARBON

WHAT WE DO :

Support the Indonesian government's commitment to addressing climate change, effective management of blue carbon ecosystems is a crucial strategy aimed at reducing emissions, enhancing economic performance, and achieving Indonesia's Nationally Determined Contribution (NDC) targets by 2030.

National Scale

DRAM :
Mitigation Action Design Document

RAM :
Mitigation Plan of Action Document

DRAM and RAM document pilots in Savu Sea, Riau Islands, Central Java and West Nusa Tenggara

Prospectus Document for Blue Carbon in Marine Protected Areas

A document that provides an overview of the potential for blue carbon in Indonesia. Together with MMAF, KI, WWF, RARE, YPL, CTC, and USAID Kolektif, we participated in the drafting of the document

Practical Guidelines for the Inventory of Greenhouse Gas Emissions in MPAs

This document can be used as a technical guide for blue carbon data collection and processing methods to produce standardized national data.

Local scale

Central Java

	Rehabilitation	Survival Rate	Nursery (Number seedlings produced)
Mangrove	Mangrove: 149,500 seeds (15 ha)	Mangrove: 79.8%	Mangrove: 89,000
Seagrass	Seagrass: 500 seeds (500 m2)	Seagrass: 50%	Seagrass: 1,500

Saleh Bay

Saleh Bay located between Sumbawa and Dompu districts, is a strategic location for the protection and management of blue carbon ecosystems. Currently, FRCI supports various activities including

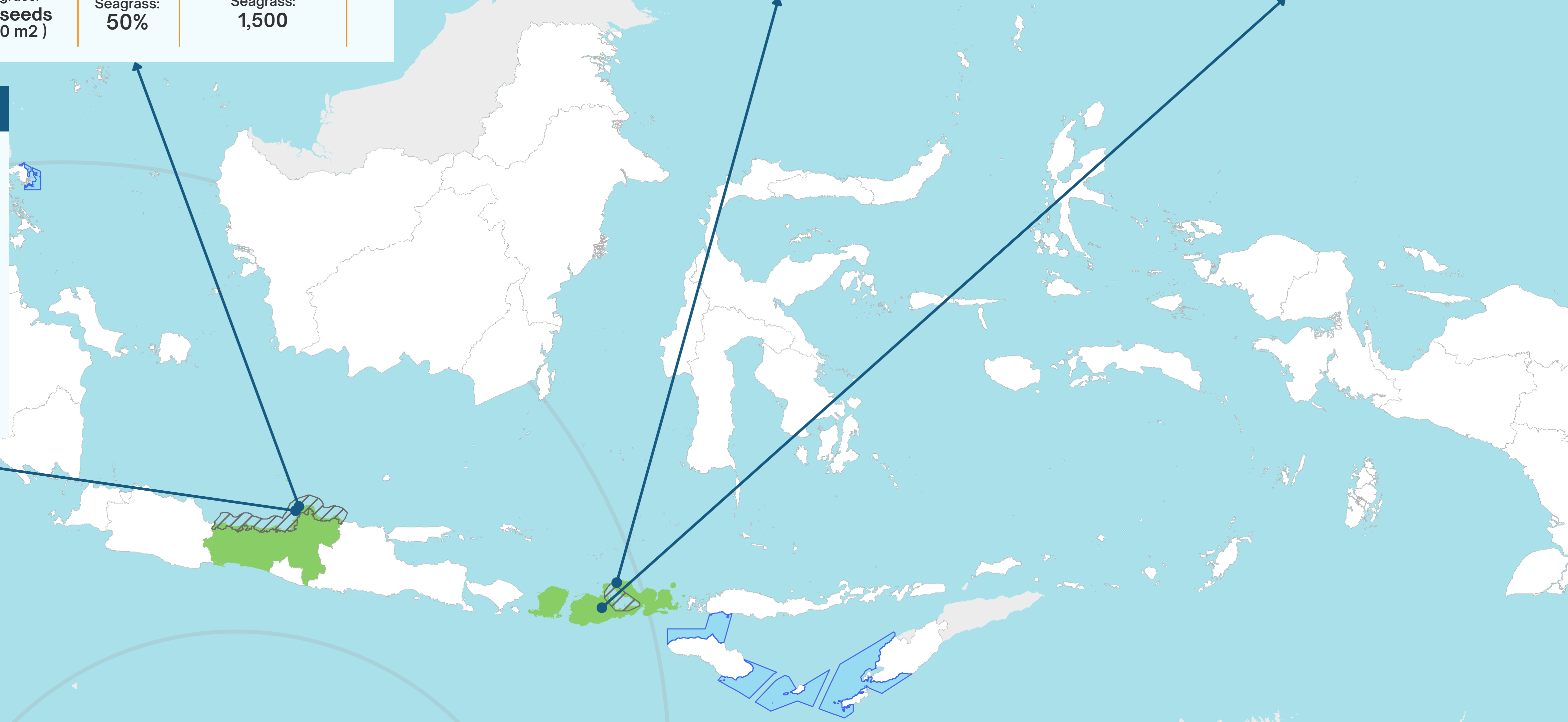
1. **Ecosystem restoration, rehabilitation, and conservation**
2. **Support for community livelihoods**
3. **Coordination with stakeholders**
4. **Blue swimming crab survey**

West Nusa Tenggara

	Rehabilitation	Nursery (Number seedlings produced)
Mangrove	Mangrove: 9,500 seeds (1 ha)	Mangrove: 9,500
Seagrass	Seagrass: 2,000 seeds (2 ha)	

Central Java

FRCI's role in **empowering coastal communities** is very significant, especially through **mangrove restoration, rehabilitation, and livelihood development** approaches based on local potential. FRCI focuses on sustainable management of natural resources, by providing technical training and assistance to communities to optimize the use of coastal resources such as fisheries, aquaculture, and ecotourism



On Central Java, our actions are :

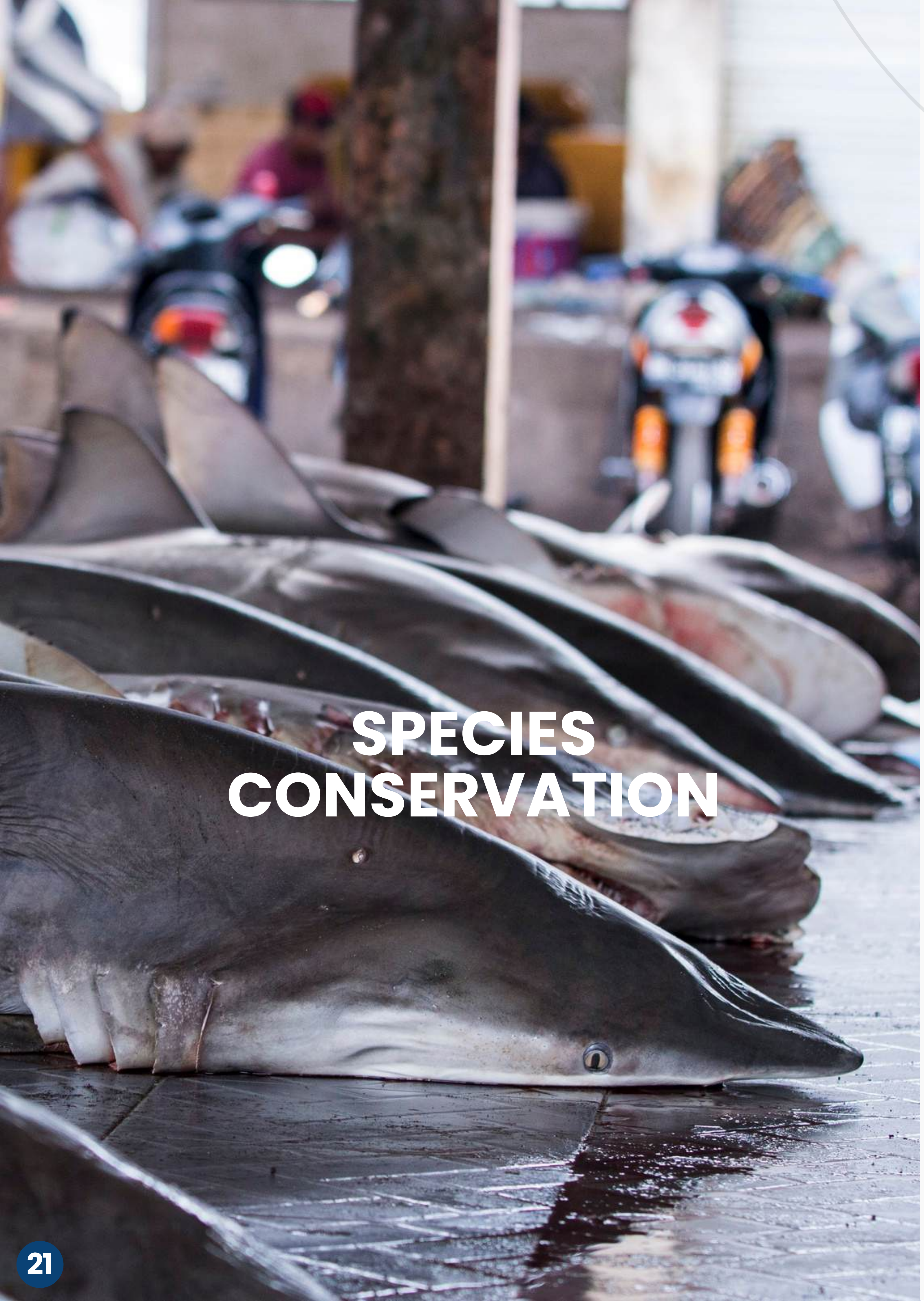
235 communities
Assisted in blue carbon rehabilitation

7 community groups
Supported in North Coast and South Coast Java

45 people
Mangrove Crab Cultivation Training

30 people
Mangrove Nursery Training

15 people
Agriculture Training



SPECIES CONSERVATION

Shark and Rays Conservation Program

Indonesia is the world's largest catcher of sharks and rays, driven by high domestic demand for food security and international markets, particularly for valuable exports like shark fins. However, these fisheries face challenges such as overexploitation, illegal trade, and inadequate management, with many species requiring strict regulation under CITES. To address these issues, our initiative focuses on building institutional capacity, enhancing CITES compliance, improving data collection for sustainable fisheries management, and engaging stakeholders to strengthen governance and conservation efforts.

What We Do

Support effective management of Indonesia's shark and ray conservation areas through research, capacity building, and policy strengthening in shark and ray conservation and sustainable fisheries.

Strengthening Indonesia's capacity to reduce illegal shark fisheries and trade

- Held **national training programs** developed for government staff to improve national capacity on shark identification
- Funded **3 research-driven PhDs**

Collected samples of shark products and genetically tested them using DNA barcoding with standardized gene regions called COI gene methods

369 **samples**

Tracing the supply chain of shark and ray meat products

Our actions

FRCI conducted a massive, nation-wide survey in the main shark-landing sites to identify the supply chain of elasmobranch meat. Working alongside local authorities, we surveyed fishers, fish auction sellers, fish market traders, and meat processing industries.

Achievements

Conducted field surveys in priority areas to assess trends of Shark and Ray Exploitation and Consumption

36	**cities**
10	**provinces**
788	**respondents**

Ocean Accounts

The Ocean Accounts (OA) Initiative in Indonesia began in 2020 through a scoping assessment, executing pilots, and developing a roadmap for national implementation.

The OA Pilot in Indonesia has been implemented in 10 National Marine Protected Areas (MPAs) and other thematic areas such as **fisheries management, blue carbon, and marine spatial planning.**

10
National Marine Protected Areas

We have fully mapped the National Coastal Ecosystem Extent of Indonesia (2018 and 2021) for the ecosystems of:


Coral reefs


Seagrass Beds


Mangroves

Ocean Accounts Projects

- OA Pilot for MPA
- OA Pilot for Blue Carbon Accounting and Marine Spatial Planning
- OA Pilot for Fisheries Accounting

OUTPUTS

OA-related Indonesian National Standards (SNI)


SNI
SNI 9258:2024
Neraca sumber daya laut – Ekosistem pesisir dan pulau-pulau kecil
BSN

SNI 9258:2024 on Ocean Accounts


SNI
SNI 9257:2024
Spesifikasi informasi geospasial – Neraca spasial habitat pesisir dan bentik laut dangkal
BSN

SNI 9257: 2024 on Geospatial Information Specifications

The OA dashboard

The dashboard is an interactive platform designed to display comprehensive Ocean Accounts (OA) information at various levels, including national, FMA, provincial, and pilot areas within Marine Protected Areas. The dashboard is periodically updated using remote sensing technology and field surveys and is being integrated with the MMAF command center and SIDAKO system to enhance data accessibility and support better ocean management.

Andalusia and System Dynamics

Andalusia is an abbreviation for the Analysis of Indonesia's Ocean Resources. The Andalusia report titled "Making Ocean Capital Visible" contains the conceptual framework, assessment methods, and the economic value of ocean resources in Indonesia.

System Dynamic is a cutting-edge computer modelling that is embedded in the ocean accounts dashboard to predict the impact of investment on ecosystem condition, and how much ecosystem rehabilitation is needed to minimize the impacts and healthy balance of the ecosystem and economy.

Training and Capacity Building

1. The OA Indonesia Development Team presented 4 (four) abstracts in the "Side event of Ocean Accounts" at the 19th Islands of the World Conference in Lombok
2. We organized the special session "Ocean accounts further the management of Marine Protected Areas aligned with the Global Biodiversity Framework" at the 5th Global Dialogue on Sustainable Ocean Development in Bali in July 2024
3. We participated in the workshop "Short-term mission to Statistics Norway (SSB) from partners of the Indonesia Oceans for Development Project Component 1"
4. We organized the "Training on Ocean Accounts 2024" for 18 MMAF staff

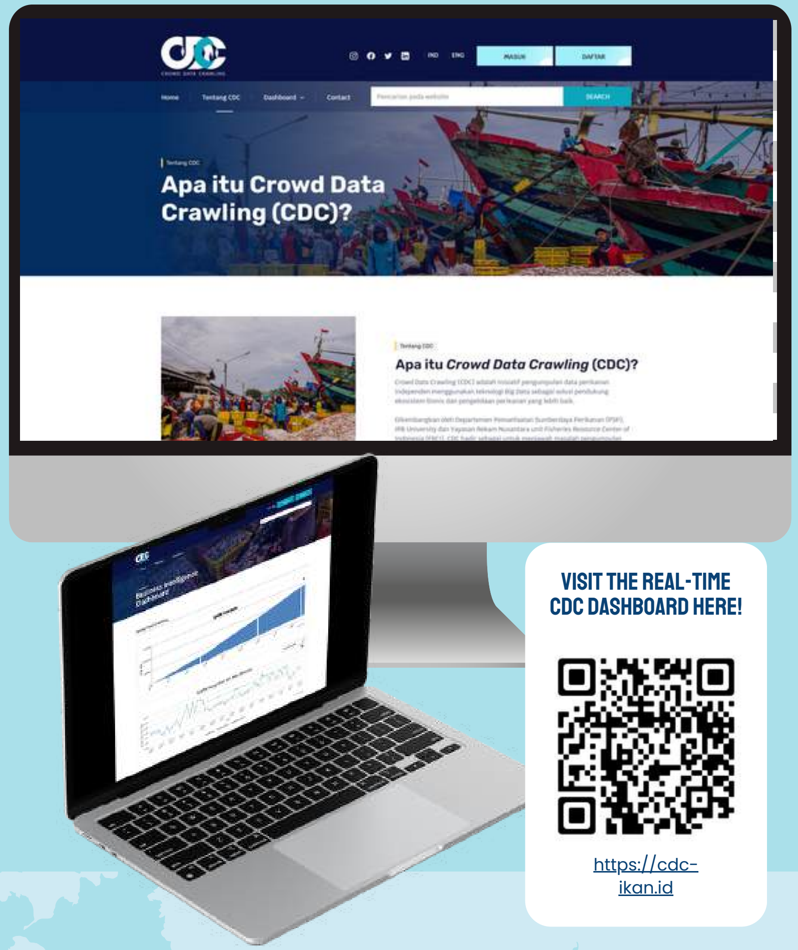
Science and Advanced Technology

We believe **technological advancements** in fisheries management in combination with **transdisciplinary approach** and **stakeholder engagement** can help ensure that management measures are ecologically appropriate and socially equitable.

Hence, we develop citizen science data collection using **IKAN mobile app**, Internet-Based Fisheries Big Data Collection Through **CDC (Crowd Data Crawling)**, and **AI (Artificial Intelligence)**-Based Fish Identification and Data Analysis to provide robust stock estimation, analysis and management recommendation.

23,090

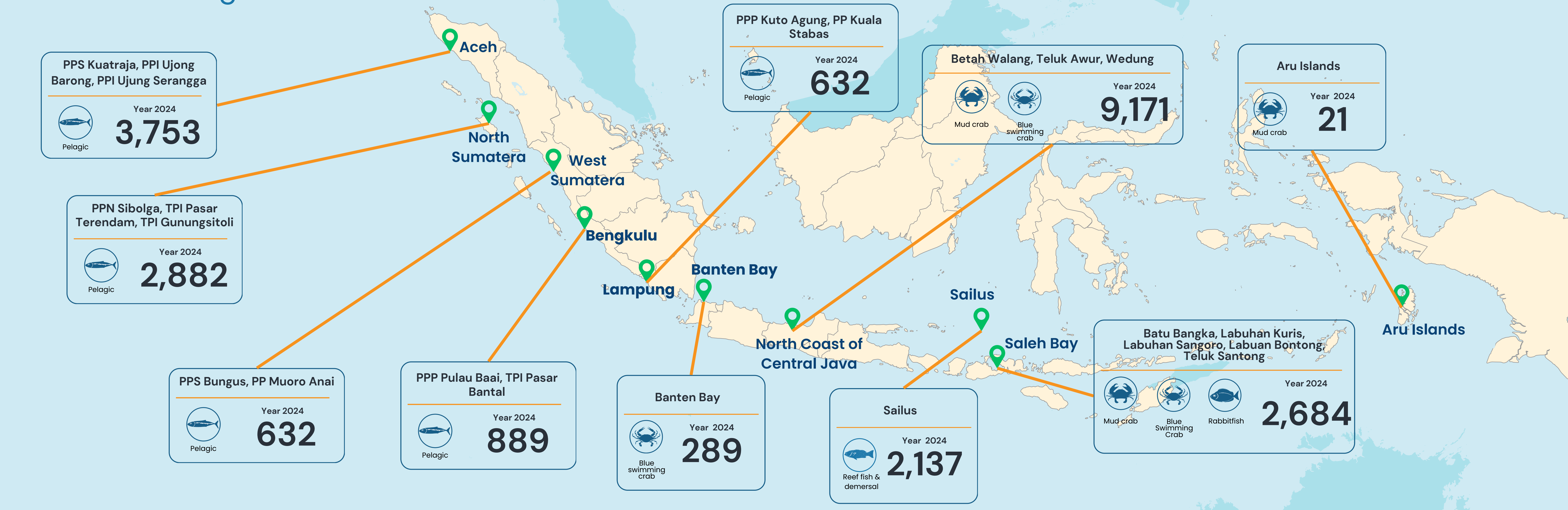
Data Collected in 2024



VISIT THE REAL-TIME CDC DASHBOARD HERE!

https://cdc-ikan.id

Number of individuals measured for fish length at each site



FRCI received the 2024 EUTECH SDGs Awards for SDG14

SFACT SCHOLARSHIP

5 awardees

Awarde research topics

1. Coastal State Line
2. Fishing Capacity
3. Shark Management
4. Non-tax State Revenue of Tuna Species
5. Shark Management in FMA 573

Marine Fisheries Technology, Faculty of Fisheries and Marine Science, IPB University

Our participation in international events

9th World Fisheries Congress with the theme "Fish and Fisheries at the Food-Water-Energy Nexus"

Seattle, Washington, March 3-7, 2024

7th International Conference of the Asian Society of Ichthyologists 2024

Penang, May 2024

The Nature-based Solutions Conference

Oxford 18-20 June 2024

Islands of the World Conference XIX "Islands and Resilience: Global Opportunities"

Lombok, West Nusa Tenggara, June 2024

the 5th Global Dialogue on Sustainable Ocean Development

Bali, Indonesia, July 2024

3rd International Conference of International Centre for Indo-Pacific Studies, India

India, July 2024

The 7th International Marine Conservation Congress (IMCC7)

Cape Town, South Africa, October 2024

Conservation Social Science Conference 2024

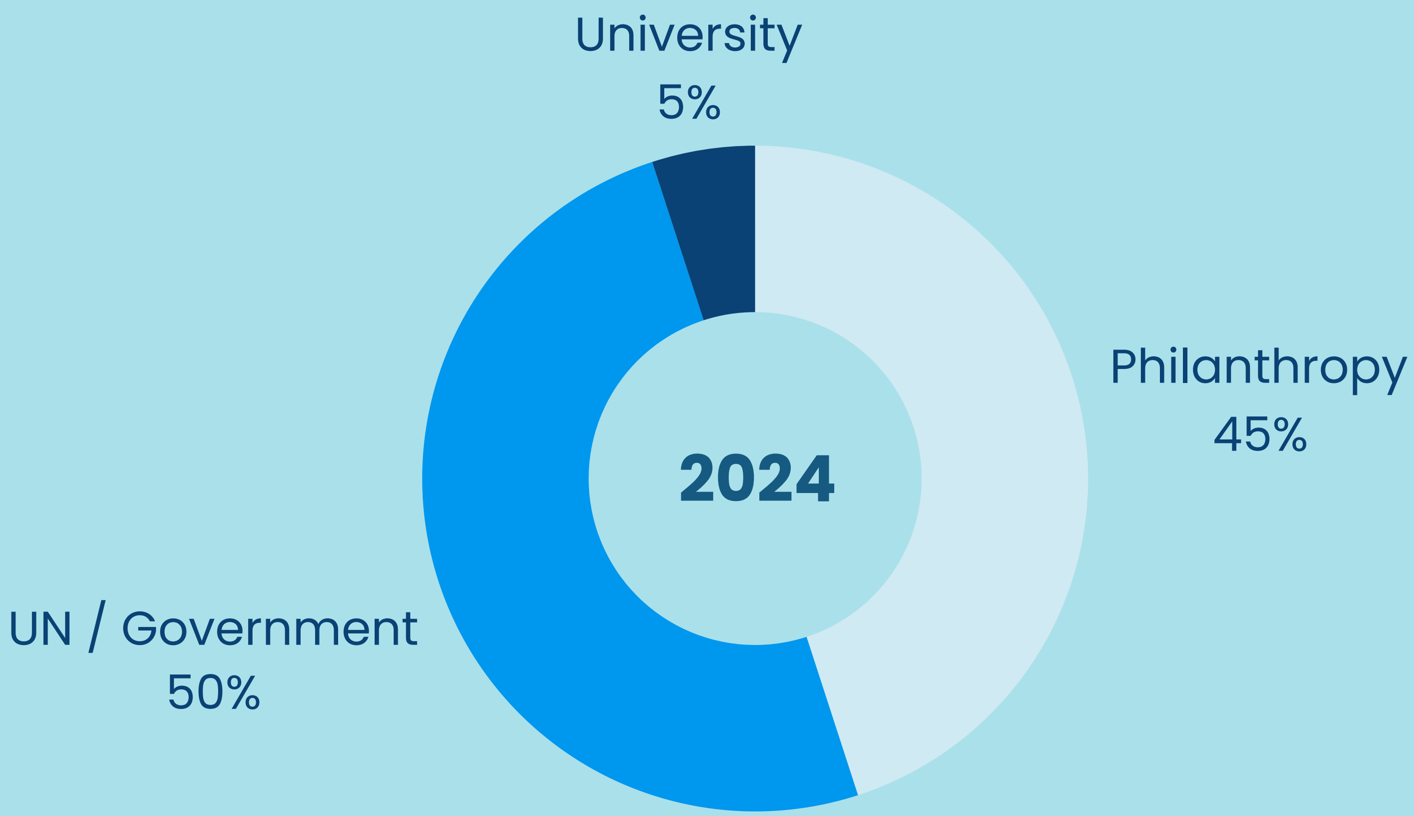
20-21 November 2024 (Online)

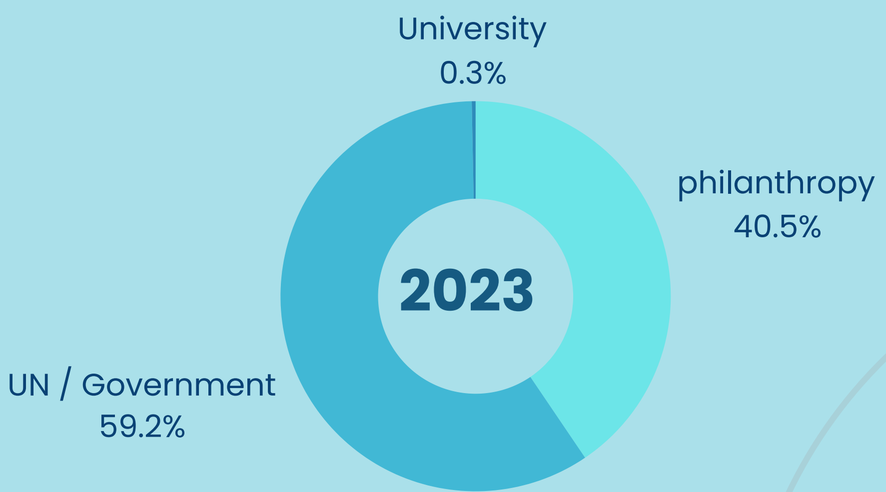
International Conference on Sustainable Coral Reefs

Manado, Indonesia,
Friday, December 13 2024 -
Sunday, December 15 2024

Finances

FRCI is grateful for the continuous support from our funders. The leverage from philanthropy, the UN, governments of both national and international, as well as universities has enabled our various work in fisheries and marine conservation.





Partners

MPA and OECM Consortium

Supporters

OUR TEAM (2024)

Dr. Irfan Yulianto	Senior for Marine and Fisheries
Dr. Heidi Retnoningtyas	Director for Ocean Program
Dr. Arisetiarso Soemodinoto	Senior Advisor
Dr. Toni Ruchimat	Senior Advisor
Prof. Dr. Ir. Budy Wiryawan	Senior Advisor
Dr. Yudi Herdiana	Advisor
Dr. Mohamad Natsir	Adjunct Researcher
Prof. Dr. Asadatun Abdullah	Principle Investigator for Sharks and Rays
Annisya Rosdiana	Manager for Ocean Accounts
Efin Muttaqin	Manager for Marine and Species Conservation
Oktavianto Prastyo Darmono	Manager for Cental Java Program
Jessica Pingkan	Manager for Blue Carbon
Dr. Adipati Rahmat Gumelar	Manager for GEF Project
Intan Destianis Hartati	Junior Manager for Sustainable Fisheries and Conservation
Yuliana Fitri Syamsuni	National Coordinator of Ocean for Development Program
Rizqi Aimmatul Maulidiyah	Marine Policy Coordinator
Ahmad Syauqi Jafani	Pelagic Fisheries Assessment Coordinator
Dr. Soraya Gigentika	West Nusa Tenggara Program Coordinator
Dr. Lilik Teguh Pambudi	Private Sector, Business and Partnership Specialist
M. Iqbal	Artificial Intelligent Development Specialist
Nabila Nur Septiani	Marine Conservation Economic Specialist
Faridatun Amalia Hasanah	GEF-6 Safeguard and Gender Specialist
Jones V. Rahanjaan	GEF-6 Site Manager 718 (Molucca Tenggara District)
Ahadar Tuhuteru	GEF-6 Knowledge Management and M&E and Communication Specialist
Yoppy Endano	GEF-6 Finance and Administration Specialist
Hasan Sangadji	GEF-6 Site Manager 715 (East Seram District)
Kuriani Imba Rondaiyantingadji	GEF-6 Site Manager 717 (Wondama Bay District)
Ayi Warmia	FRCI Program Officer
Marsha Hamidah	FRCI Program Officer
Willy Puspa Irawan	FRCI Program Officer
Agavia Kori Rahayu	FRCI Program Officer
Nurul W. Q. Manik	FRCI Program Officer
Siti Zanuba Aisyah	FRCI Program Officer
Wahyu Putri Fajar Rahmalinda	FRCI Program Officer
Fiki Hidayati	Community Assistant for Central Java Program
Rahman Firdaus	Community Officer
Ibnu Rusdi	FRCI IT Officer
Rahadianti	Program Support Officer
Novitasari	Assistant of Program Support Officer
Tantry Kusumahati	Finance Officer
Muhamad Firmansyah	Finance Assistant
Nuranti Anarkhis	Project Secretary

FRCI
FISHERIES RESOURCE CENTER OF INDONESIA

IMPACT REPORT FISHERIES RESOURCE CENTER OF INDONESIA 2024

EDITORS
Siti Zanuba Aisyah, Agavia Kori Rahayu

CONTRIBUTORS
Heidi Retnoningtyas, Annisya Rosdiana, Efin Muttaqin, Mohamad Natsir, Intan Destianis Hartati, Oktavianto P. Darmono, Jessica Pingkan, Neni Nuraeni

LAYOUT
Mas'ud Wijaya

ADDITIONAL ILLUSTRATIONS
Pandu Darma Wijaya, Agavia Kori Rahayu